

SIT, Holography, and the AdS / CFT Translation Split

A Unit-Bearing Time-Density Probe and a First-Order Extremal-Surface Shift in Regulated AdS3

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Abstract

Draft 6 of this paper ended with a precise open obligation: replace a verbal comparison between a local time-density field and holographic entropy with an explicit object, boundary deformation, entropy functional, and extremal-surface calculation. This successor discharges that obligation in one deliberately narrow setting. We use a regulated, time-reflection-symmetric interval in Euclidean Poincare AdS3. A dimensionless scalar profile

$$\chi_1 = (\tau^2 + x^2 + z^2 - a^2)/(2 a z)$$

solves the free bulk equation with $m^2 L^2 = 3$, vanishes on the undeformed Ryu-Takayanagi geodesic, and has a nonzero proper-normal derivative there. It is embedded into the unit-bearing time-density probe

$$\rho_t(\epsilon) = \rho_{t0} + \epsilon \omega_* \chi_1,$$

where both ρ_{t0} and ω_* have units of inverse time. Its leading boundary coefficient is the length-valued source $J_1 = (\tau^2 + x^2 - a^2)/(2a)$ for a dimension-three scalar operator.

To make the quantum-extremal-surface question executable, we stipulate a local first-order response of the bulk-entropy term and extremize the resulting generalized entropy with fixed regulated endpoints. The analytic displacement of the curve is

$$n_\epsilon(s)$$

$$= -4 \epsilon \alpha \ell_P$$

$$[1 - \cosh(s/L)/\cosh(S/L)]$$

$$+ O(\epsilon^2),$$

$$\cosh(S/L) = a/\delta.$$

The Dirichlet Hessian is positive, the result is reproduced by an independent finite-difference solve, and the standard regulated RT surface is recovered exactly at zero deformation. A counterexample shows that a nonzero field whose normal profile is even about the surface produces no first-order shift. The earned result is therefore a conditional QES-shift theorem for the stated toy functional, not a general statement that time-density deformations move extremal surfaces.

Scope and evidence boundary

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Claim ceiling This note proves a conditional perturbative result in one regulated toy geometry. It does not give a complete AdS / CFT dictionary, derive a bulk von Neumann entropy from a quantum state, validate SIT empirically, or establish a complete holographic theory.

1. The question this draft actually answers

The source-level intuition is simple: suppose a model treats the local number of clock increments per coordinate time as a field rather than a fixed background. Can one place a small version of that field in a standard holographic geometry, identify a dimensionally consistent boundary source, and compute whether an entropy-extremizing curve moves?

There are several much larger questions nearby. Does SIT define a consistent quantum gravity? Is there a microscopic CFT dual? Can one reconstruct the bulk from boundary data? Can a time-density variable explain black-hole entropy or information recovery? This draft answers none of them. It asks only for a controlled local calculation in a named background.

The result has five parts:

1. a scalar bulk profile with an exact Klein-Gordon check;
2. an inverse-time physical probe built from that dimensionless profile;
3. a standard source coefficient with a closed unit ledger;
4. a stipulated generalized-entropy response with a positive shape Hessian;
5. an analytic first-order displacement with an exact zero-source limit.

That is enough to turn one Draft 6 obligation into mathematics. It is not enough to promote the larger correspondence language.

2. Relation to Draft 6 and the public record

Draft 6 is preserved byte for byte. It remains the canonical manuscript in the existing control materials. Its central guardrail was that SIT could be translated against holographic vocabulary while the package did not yet supply an AdS / CFT dictionary, an RT derivation, or a QES derivation.

This successor does not silently weaken that guardrail. Instead, it moves one item from “missing calculation” to “conditional toy calculation.” The distinction matters:

- Draft 6 said that an entropy functional and extremization rule were needed.
- Draft 7 writes one such functional and solves its linearized extremality equation.
- The bulk-entropy term in that functional is still stipulated rather than derived from a microscopic state.
- Therefore the result is not a derivation of the standard QES prescription from SIT.

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3. Standard holographic baseline

The calculation uses only a small standard backbone.

Maldacena’s original correspondence proposal and the Gubser-Klebanov-Polyakov and Witten prescriptions establish the familiar source / field framework in controlled AdS / CFT examples: boundary values of bulk fields act as sources for boundary operators, with the scalar mass tied to the operator dimension. Nothing in that statement licenses an arbitrary bulk model as a new duality. Here it is used only to assign a conventional scalar source and its units.

Ryu and Takayanagi give the classical area prescription for holographic entanglement entropy. For a single interval in Poincare AdS₃, the relevant constant-time geodesic is a semicircle. Faulkner, Lewkowycz, and Maldacena identify the leading bulk-entanglement correction. Engelhardt and Wall define a quantum extremal surface by extremizing generalized entropy, including area, bulk entropy, and counterterm structure.

This draft follows that order:

standard interval geometry

- > standard scalar asymptotics
- > explicit source deformation
- > explicit generalized-entropy toy response
- > extremality equation
- > displacement and limit checks.

The standard references do not supply the local entropy response chosen in Section 7. That term is the new assumption being tested, and it remains visibly separated from the standard ingredients.

4. One geometry and one regulator

Take Euclidean Poincare AdS3 with radius L,

$$ds^2 = (L^2/z^2) (d\tau^2 + dx^2 + dz^2), \quad z > 0.$$

Consider an interval centered at $x = 0$ on the reflection slice $\tau = 0$. The radial cutoff is $z = \delta$, with $0 < \delta < a$. Instead of sending the endpoints to the conformal boundary, fix the regulated endpoints on the classical curve

$$\gamma_0: \tau = 0, \quad x^2 + z^2 = a^2.$$

A proper-arclength parameterization is

$$\begin{aligned} x(s) &= a \tanh(s/L), \\ z(s) &= a \operatorname{sech}(s/L), \\ -S &\leq s \leq S, \\ \cosh(S/L) &= a/\delta. \end{aligned}$$

Direct substitution gives $x(s)^2 + z(s)^2 = a^2$. The induced line element is ds , so the regulated geodesic length is $2S$.

Introduce a signed proper normal $n(s)$ within the $\tau = 0$ hyperbolic slice. Positive n points toward $x^2 + z^2 > a^2$. In Fermi coordinates around the geodesic, the slice metric is

$$ds_{\text{slice}}^2 = dn^2 + \cosh^2(n/L) ds^2.$$

We impose $n(-S) = n(S) = 0$. This fixed-endpoint regulator makes the inverse of the shape operator unambiguous. It also means that the result should not be read as a regulator-independent CFT observable before the appropriate counterterm and endpoint analysis is supplied.

Reflection symmetry suppresses the second, time-directed normal in this toy problem. The calculation is not a proof of full Lorentzian extremality under arbitrary codimension-two deformations.

5. A unit-bearing time-density probe

The private SIT and SDT source genealogy uses ρ_t for a local number of time increments per coordinate time. Its stated dimension is inverse time. To preserve that intent while using a standard scalar equation, separate the physical units from the normalized shape:

$$\begin{aligned} \rho_t^\sim(\epsilon)(\tau, x, z) \\ = \rho_{t0} + \epsilon \omega_* \chi_1(\tau, x, z). \end{aligned}$$

Here:

$[\text{rho_t0}] = [\text{omega_}] = \text{inverse time},$
 $[\text{epsilon}] = [\text{chi_1}] = 1.$

The profile chosen for the one worked example is

$$\text{chi_1}(\tau, x, z) \\ = (\tau^2 + x^2 + z^2 - a^2)/(2 a z).$$

For the Poincare metric, the scalar Laplacian is

$$\text{nabla}^2 \text{chi} \\ = [z^2(\text{chi_tautau} + \text{chi_xx} + \text{chi_zz}) - z \text{chi_z}]/L^2.$$

Symbolic differentiation gives

$$(\text{nabla}^2 - 3/L^2) \text{chi_1} = 0.$$

Thus the normalized profile is a free scalar solution with $m^2 L^2 = 3$. This exact equation is useful but modest: it establishes the kinematics of the probe, not a full interacting bulk action.

On the undeformed geodesic,

$$\text{chi_1}(0, x, z) = 0 \text{ when } x^2 + z^2 = a^2.$$

In the signed Fermi normal,

$$\text{chi_1} = \sinh(n/L), \\ \text{partial_n chi_1 at } n=0 = 1/L.$$

The field value is therefore zero on the original curve while its normal gradient is nonzero. That makes it a clean first-order shape probe. If the scalar is expanded around zero amplitude, its ordinary stress tensor begins quadratically in ϵ ; treating the metric as fixed is consistent through the first-order calculation performed here.

This last statement has a scope condition: it assumes no linear tadpole, nonzero background scalar, or explicit order- ϵ metric coupling. A more complete SIT action could violate that assumption and would need a joint metric-scalar calculation.

6. Boundary source and dimension check

For a scalar in AdS with $d = 2$ boundary dimensions, the standard relation is

$$\Delta(\Delta-d) = m^2 L^2.$$

With $m^2 L^2 = 3$, the larger root is $\Delta = 3$. Near $z = 0$,

$$\text{chi_1} \\ = z^{(-1)} J_1(\tau, x) + O(z),$$

$$J_1(\tau, x) \\ = (\tau^2 + x^2 - a^2)/(2a).$$

The corresponding conventional source insertion is

$$I_{\text{boundary}}[\epsilon] \\ = I_{\text{CFT}} + \epsilon \int d\tau dx J_1 O_3.$$

The dimensions close:

```
[d tau dx] = length^2,
[J_1] = length,
[0_3] = length^(-3),
[epsilon] = 1.
```

Therefore the added action is dimensionless. The inverse-time units of ρ_t reside in ω_* ; they are not incorrectly assigned to the AdS scalar profile or to the boundary source.

This is a field / source pair, not a complete dictionary. A complete dictionary would additionally require, among other things, the full bulk action, normalization of correlators, boundary conditions and counterterms, identification of the boundary theory and state, interaction terms, symmetry matching, and a reconstruction domain.

The polynomial source grows on the noncompact boundary plane. In this note it is defined inside the finite Dirichlet regulator used for the interval, or equivalently multiplied by a smooth compact support that agrees with the displayed profile on the region being probed. No normalizable global state on the full plane is constructed.

7. A stipulated generalized-entropy response

Write entropy in dimensionless units S/k_B and define

```
S_gen[n;epsilon]
= (1/(4 ell_P)) integral_-S^S ds
  sqrt((dn/ds)^2 + cosh^2(n/L))
+ S_bulk^0
+ epsilon alpha integral_-S^S (ds/L) sinh(n/L).
```

The first term is the length contribution for a curve $n(s)$ in the Fermi metric. The constant $\text{ell}_P = G_3 \hbar / c^3$ has units of length. The term S_{bulk}^0 is taken to be shape independent within this local exercise.

The final term is the decisive assumption:

```
delta S_bulk^toy[n]
= epsilon alpha integral (ds/L) sinh(n/L).
```

It is a dimensionless, local first-order response whose integrand uses the same normalized profile χ_1 . The coefficient α is dimensionless. This form is useful because it is odd in the signed normal and therefore supplies a nonzero linear shape source at $n = 0$.

It must not be confused with a derived bulk von Neumann entropy. Standard bulk entanglement entropy is generally nonlocal and depends on a state and a subregion. A replica derivation, modular-Hamiltonian calculation, or explicit density matrix could select a different response or no response. The theorem below is conditional on this displayed ansatz.

Expanding at small n/L gives

```
S_gen
= constant
+ (1/(8 ell_P)) integral_-S^S ds
  [(n')^2 + n^2/L^2]
+ epsilon alpha integral_-S^S ds n/L^2
+ 0(n^3/L^3, epsilon n^3/L^3).
```

The undeformed area functional has no linear term because γ_0 is a geodesic. The bulk response supplies the only order- ϵ forcing term in this fixed-background model.

8. First-order displacement theorem

Proposition

Let L, a, δ , and ℓ_P be positive with $\delta < a$. Fix the regulated endpoints $n(-S) = n(S) = 0$, where $S = L \operatorname{arcosh}(a/\delta)$. Assume the generalized-entropy toy functional in Section 7 and the perturbative regime

$$4 |\epsilon| \alpha \ell_P (1-\delta/a)/L \ll 1.$$

Then the reflection-symmetric extremal curve has the unique first-order normal displacement

$$\begin{aligned} n_\epsilon(s) &= -4 \epsilon \alpha \ell_P \\ &\quad [1 - \cosh(s/L)/\cosh(S/L)] \\ &\quad + O(\epsilon^2). \end{aligned}$$

Derivation

Varying the quadratic functional yields

$$(1/(4 \ell_P))(-n'' + n/L^2) + \epsilon \alpha/L^2 = 0.$$

Equivalently,

$$\begin{aligned} (-d^2/ds^2 + 1/L^2)n &= -4 \epsilon \alpha \ell_P/L^2. \end{aligned}$$

The general solution is a constant particular solution plus $A \cosh(s/L) + B \sinh(s/L)$. Reflection symmetry sets $B = 0$. The two fixed endpoint conditions then determine A and give the displayed result.

The Dirichlet Hessian is

$$H = -d^2/ds^2 + 1/L^2.$$

Its eigenvalues on $[-S, S]$ are

$$\lambda_k = 1/L^2 + (\pi k/(2S))^2, \quad k = 1, 2, \dots$$

and are strictly positive. The linearized extremum is therefore unique and is a local minimum of the quadratic functional.

At the apex $s = 0$,

$$\begin{aligned} \delta n_* &= -4 \epsilon \alpha \ell_P (1-\delta/a) \\ &\quad + O(\epsilon^2). \end{aligned}$$

At $x = 0$, proper normal and Poincare radial displacement satisfy $\delta z = (a/L) \delta n$, so

$$\begin{aligned} \delta z_* &= -4 \epsilon \alpha \ell_P (a-\delta)/L \\ &\quad + O(\epsilon^2). \end{aligned}$$

With the chosen normal orientation, positive ϵ moves the apex toward smaller z , namely toward the boundary. Reversing the sign moves it deeper into the bulk.

The position changes at first order while the extremized entropy changes only at second order in this construction: the undeformed surface is stationary, and the stipulated response itself vanishes at $n = 0$. This is not a contradiction. A stationary point can move linearly while the optimum value begins quadratically.

9. Standard zero-deformation limit

Set $\epsilon = 0$. The boundary source insertion vanishes, the time-density probe returns to ρ_{t0} , and the generalized entropy becomes

$$\begin{aligned} S_{\text{gen}}[n;0] &= (1/(4 \ell_P)) \int ds \\ &\quad \sqrt{(n')^2 + \cosh^2(n/L)} \\ &\quad + S_{\text{bulk}}. \end{aligned}$$

The unique small solution of the Dirichlet extremality equation is $n_0(s) = 0$. Thus the original semicircle is recovered exactly. Its regulated area contribution is

$$S_{\text{area}}[\gamma_0] = 2S/(4 \ell_P).$$

The analytic shift also vanishes when $\alpha = 0$ or in the probe classical limit $\ell_P \rightarrow 0$.

This limit is an acceptance gate rather than a decorative observation. Any successor model that failed to return the standard RT curve after removing its deformation would not be a controlled perturbation of the stated baseline.

10. A counterexample to the overbroad claim

It would be false to conclude that every nonzero time-density deformation causes a first-order QES shift. Consider instead the even normal profile

$$\chi_{\text{even}}(n) = \cosh(n/L) - 1.$$

This profile is nonzero away from the surface but satisfies

$$\begin{aligned} \chi_{\text{even}}(0) &= 0, \\ \partial_n \chi_{\text{even}} \text{ at } n=0 &= 0. \end{aligned}$$

If the local entropy response were proportional to χ_{even} , its expansion would start at n^2 . It would change the Hessian but supply no linear forcing term. The first-order displacement would be zero.

The discriminating statement is therefore:

Within the fixed-endpoint toy functional, the first-order QES displacement is controlled by the normal derivative of the stipulated entropy response on the baseline surface, not merely by the presence of a nonzero bulk field.

The same logic identifies two further failure modes. If the Dirichlet Hessian has a zero mode, the inverse-shape-operator formula is not available without a collective-coordinate treatment. If endpoint motion is allowed, boundary terms change the variational problem.

11. Executable checks

The companion model performs exact symbolic and independent numerical checks. The checked identities are:

- the Klein-Gordon residual is zero;
- the arclength curve lies on $x^2 + z^2 = a^2$;
- χ_1 vanishes on that curve;
- $z \chi_1$ tends to the stated boundary source;
- the proper-normal derivative is $1/L$;
- the analytic displacement satisfies the ODE;
- both fixed endpoint conditions hold;
- the zero-deformation profile is exactly zero;
- the Dirichlet Hessian is positive;
- a finite-difference solve agrees with the analytic solution;
- the displacement sign matches the stated orientation;
- a numerical fixture satisfies the perturbative bound;
- all displayed dimensions close;
- the even-profile counterexample has zero linear derivative.

For the stored fixture,

```
L = 2
a = 5
delta = 0.25
ell_P = 0.01
epsilon = 0.05
alpha = 0.4
```

the generated model/results.json is the acceptance authority for the exact run. The script and unit tests are intentionally separate: the symbolic run checks the equations, while the finite-difference solve checks that the analytic boundary-value solution is not merely reasserted.

These are model checks, not empirical tests. They establish internal consistency of the bounded calculation.

12. Source-faithful vocabulary

The private source genealogy contains ambitious language about local time-density, bulk / boundary effects, and a holography-focused paper. This draft preserves the narrow, recurring intent and rejects unsupported promotion.

The plain-language sequence is:

1. The proposed physical quantity is a local count rate of clock increments.
2. In a standard AdS setup, a normalized scalar shape can be assigned a conventional boundary source.
3. A specified entropy response acts as a source in the surface-shape equation.
4. Only then do we call the normalized construction a time-density probe.
5. The discriminating check is whether its normal entropy derivative is nonzero; the even-profile counterexample shows why that condition matters.

No new branded term is needed. Removing “time-density probe” from the calculation leaves every equation and inference understandable. The phrase is therefore descriptive, not load-bearing.

The source texts also contain stronger physical and empirical assertions. Those assertions are not imported into this draft. The inverse-time unit of ρ_t and the intent to study a controlled entropy or QES shift are treated as authorial design inputs. The standard holographic equations and the conditional theorem are established independently in the displayed model.

13. What remains open

The central open problem is microscopic entropy. To turn the yellow conditional result into a stronger holographic claim, one would need an explicit bulk quantum state or density matrix and a controlled derivation of δS_{bulk} under the scalar source. Plausible routes include a first-law of entanglement calculation, a modular-Hamiltonian response in a specified state, or a replica computation with the required counterterms. Any of those routes may change the local ansatz used here.

Other open obligations are:

- specify a complete bulk action and its variational boundary terms;
- identify a boundary theory rather than only an operator dimension;
- perform holographic renormalization for the irrelevant source;
- replace the polynomial source with a globally controlled preparation;
- include possible order- ϵ metric response if the full action permits it;
- vary both codimension-two normals in Lorentzian signature;
- show state, symmetry, and normalization continuity across the map;
- separate regulator motion from a physical surface displacement;
- subject the calculation to independent expert review;
- define an empirical observable before making any validation claim.

The next best mathematical leaf is not a black-hole application. It is the smaller task of choosing a specified free or weakly coupled bulk state and deriving the sign and kernel of its entropy response to the displayed scalar source. That would directly test the only new assumption used in this note.

Final-preprint endpoint decision

This paper closes at the **conditional toy-model endpoint**. The displayed bulk-entropy response remains a transparent stipulation inside one regulated generalized-entropy functional; it is not reclassified as a microscopic entropy derivation. The theorem, source deformation, inverse-time time-density field, first-order surface shift, zero-deformation recovery, and even-profile counterexample are retained without weakening.

Microscopic entropy is therefore a successor research program rather than unfinished local work for this preprint. A future paper may replace the stipulated response with a state-specific derivation and then test whether the present sign, kernel, and displacement survive. Until that work exists, this paper makes no complete-QES, complete-AdS / CFT-dictionary, black-hole- information, or empirical-validation claim.

This disposition closes manuscript scope only. Independent specialist review, Micah Blumberg's approval, provider execution, and any release freeze remain external nonblocking gates and are not authorized by this candidate.

14. Conclusion

This private successor advances the paper by one bounded rung. It places an inverse-time time-density perturbation into one standard AdS3 interval model, extracts a dimensionally consistent boundary source, stipulates a transparent bulk-entropy response, and solves the resulting first-order extremal-surface problem. The analytic result has a positive Hessian, an independent numerical check, a standard zero-deformation limit, and a counterexample that blocks an overbroad generalization.

The calculation earns exactly this statement:

For the regulated Poincare AdS3 interval and the displayed local generalized-entropy response, the unit-bearing scalar deformation produces the stated unique first-order normal shift.

It does not earn a complete AdS / CFT dictionary, a microscopic QES derivation, an empirical confirmation of SIT, or release readiness. Those boundaries are part of the result.

Appendix A. Compact unit ledger

quantity	dimension
L, a, delta, S, s, n	length
ell_P	length
tau, x, z	length
epsilon, alpha, chi_1	1
rho_t, rho_t0, omega_*	inverse time
J_1	length
O_3	inverse length^3
d tau dx	length^2
S_gen	1

Appendix B. Reproduction

From the private successor folder:

```
python -B model/run_checks.py
python -B -m unittest discover -s model -p "test_*.py" -v
```

The symbolic checks require Python and SymPy. The finite-difference boundary-value solve uses only the Python standard library.

Primary references

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4. S. Ryu and T. Takayanagi, “Holographic Derivation of Entanglement Entropy from AdS / CFT,” arXiv:hep-th/0603001.

5. S. Ryu and T. Takayanagi, “Aspects of Holographic Entanglement Entropy,” arXiv:hep-th/0605073.
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